inefficient by around 30% for smaller ones and up to 70-75% for larger ones.

Their advantages:

• Universal motors are cheaper due to their simple construction and do not require permanent magnets.

• Universal motors provide good torque at low speeds.

• These can rotate at high speed due to the connection of armature windings and field windings in series.

• Due to their small size and light weight, universal motors are preferred in many applications.

• Eddy current losses are reduced due to the presence of an electromagnetic field and heating is avoided.

• Universal motors require low power input to operate the motor.

• The engines can run at any given speed.

Their disadvantages:

• Universal motors produce relatively loud noise, which increases with motor speed.

• There are high chances of vibrations, which can damage the engine.

• There is no possibility of turning the motor in the opposite direction.

• These engines cause overheating.

• Universal motors require frequent maintenance due to rapid wear and tear of brushes and commutators.

Their applications:

• Domestic appliances like vacuum cleaners and sewing machines

• Blenders, portable drills, etc

• Polishers and other machines

• Electric shavers

AC induction motors

AC induction motors are primarily the workhorse of the industry and for good reasons. These combine performance with simplicity. The motors can be controlled directly from AC sources, which was an important function in the past. Today, thanks to digital signal processing and advanced power electronics, it is relatively inexpensive to change the speed of instant messaging.

Modern control algorithms allow induction motors to serve where expensive brush motors were previously used. The resulting energy savings directly lead to increasingly significant financial and environmental benefits. That is why controlled induction motors can be easily found

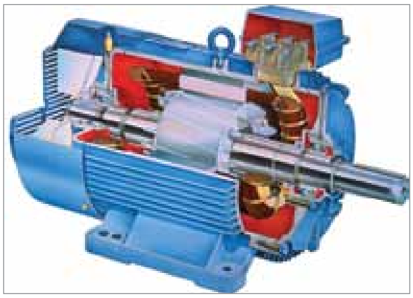

Fig. 2: An AC induction motor

in a variety of cost-critical applications such as washing machines and electric vehicles, as well as in the industrial world of blowers, pumps, fans, and material handling.

An induction motor generates torque by the interaction of its rotating stator field with induced currents in the rotor conductors. However, conductors heat the rotor, which reduces the efficiency and life of the bearings. Replacing traditional aluminium rotor conductors with copper versions helps a little, but they are expensive and can prohibit direct starting.

Their stator's electrical time constant is long; therefore, to ensure an acceptably fast response to changes in load or speed, it is common to operate the motor at a constant magnetic flux up to 'base' speed. Unfortunately, associated magnetisation losses are then present, no matter how hard the motor is running, resulting in low light-load efficiency.

Automatic stator flux reduction at low torques is possible, but only if fast control responses are not required. At speeds above base speed, the stator field weakens due to the limited supply voltage, so controlled induction motors generally provide constant mechanical energy. Efficiency drops quickly, however, because stray inductance causes the rotor current to fall behind the rotating field, so more rotor current is needed for a given torque.

Consequently, controlled induction motors are limited to a constant power speed range (CPSR) of about 2:1. Applications that require a wider CPSR, such as vehicle traction and machine tools, can be achieved by decreasing the number of winding curves and discarding excess torque capacity at lower speeds. Higher stator currents make drives more expensive and less efficient.

Please note, induction motors' nameplate efficiencies are quoted for pure sine wave operation. In real-world scenarios, inverters deliver pulses that only approximate sinusoidal current. Design teams should note that the overall system efficiency (motor and drive) will be less than the product of the individually quoted values for the motor and drive. Higher carrier frequencies help but increasing motor power results in higher drive losses.

A solution lies in placing filters between the inverter and the motor, but it introduces additional energy losses and is expensive. Also, the filters are quite big. Another disadvantage of AC induction motors is that the stator windings are distributed over many grooves in the stator core. That means long end curves that waste space and energy.

Their advantages:

• Induction motors do not require brushes, slip rings, and commutator arrangements. That means they are very simple to build.

• As there is no brush and slip ring arrangement, the mechanical or friction loss is very low, therefore, the efficiency of the induction motor is very high. The efficiency of an induction motor can be expected to be between 85-95%.

• No special arrangement is needed for the winding to create north poles and south poles like a DC motor. It automatically poles with the polarity of AC power.

• As induction motor does not have brushes, slip rings, commutator, it is a very low-cost machine compared to a DC motor or other motors.

• Although a single-phase induction motor requires a starting arrangement such as a capacitor, three-phase induction motors are auto-starting.

• Induction motors are of very simple construction, so their maintenance is much less compared to other motors.

• As an induction motor operates on AC power, there is a very low loss of I2R compared to DC motors.

• The induction motor provides very good speed range and can be operated in any environmental condition; even at 400°C.

• The starting method of induction motors is very simple. This means that induction motor starters are much sim-